The Inspection Process

The main goal of the inspection process is to detect defects in work products. Software inspections were first proposed by Fagan. Earlier inspections were focused on code, but over the years its use has spread to other work products too. In other words, the inspection process is used throughout the development process. Software inspections are now a recognized industry best practice with considerable data to support that they help in improving quality and also improve productivity.

An inspection is a review of a software work product by a group of peers following a clearly defined process. The basic goal of inspections is to improve the quality of the work product by finding defects. However, inspections also improve productivity by finding defects early and in a cost effective manner. Some of the characteristics of inspections are:

- An inspection is conducted by technical people for technical people
- It is a structured process with defined roles for the participants
- The focus is on identifying problems, not resolving them
- The review data is recorded and used for monitoring the effectiveness of the inspection process

As inspections are performed by a group of people, they can be applied to any work product, something that cannot be done with testing. The main advantage of this is that defects introduced in work products of the early parts of the life cycle, or in the work products produced by other processes like the project management process or the CM process, can be detected in that work product itself, thereby not incurring the much higher cost of detecting defects in later stages.

Inspections are performed by a team of reviewers (or inspectors) including the author, with one of them being the moderator. The moderator has the overall responsibility to ensure that the review is done in a proper manner and all steps in the review process are followed. Most methods for inspections are similar with minor variations.

The different stages in this process are: planning, preparation and overview, group review meeting, and rework and follow-up. These stages are generally executed in a linear order. We discuss each of these phases now.

Planning

The objective of the planning phase is to prepare for inspection. The author of the work product ensures that the work product is ready for inspection. The moderator checks that the entry criteria are satisfied by the work product. The entry criteria for different work products will be different. For example, for code an entry criteria is that the code compiles correctly and the available static analysis tools have been applied. The review (inspection) team is also formed in this phase. The package that needs to be distributed to the review team is prepared. The package includes the work product to be reviewed, the specifications for that work product, relevant checklists and standards. The specifications for the work product are frequently the output of the previous phase and are needed to check the correctness of the current work product. For example, when a high level design has to be reviewed, then the package must include the requirement specification also, without which checking the correctness of design may not be possible.

Overview and Preparation

In this phase the package for review is given to the reviewers. The moderator may arrange an opening meeting, if needed, in which the author may provide a brief overview of the product and any special areas that need to be looked at carefully. The objective and overview of the inspection process might

also be given in this meeting. The meeting is optional and can be omitted. In that case, the moderator provides a copy of the group review package to the reviewers.

Project name and code :
Work product name and ID:
Reviewer name:
Effort spent for preparation (hrs):
Defect List:

Sl	Location	Description	Criticality / Seriousness

Figure 2.13: Self review log.

The main task in this phase is for each reviewer to do a self-review of the work product. During the self-review, a reviewer goes through the entire work product and logs all the potential defects he or she finds in the self-preparation log. Often the reviewers will mark the defect on the work product itself. The reviewers also record the time they spent in the self-review. A standard form may be used for the self-preparation log. Relevant check lists, guidelines, and standards may be used while reviewing. Checklists specifying the type of defects to look for are particularly useful. Ideally, the self review should be done in one continuous time span. The recommended time is less than two hours—that is, the work product is small enough that it can be fully examined in less than two hours. This phase of the review process ends when all reviewers have properly performed their self review and filled the self-review logs.

Group Review Meeting

The basic purpose of the group review meeting is to come up with the final defect list, based on the initial list of defects reported by the reviewers and the new ones found during the discussion in the meeting. The entry criterion for this step is that the moderator is satisfied that all the reviewers are ready for the meeting. The main outputs of this phase are the defect log and the defect summary report. The moderator first checks to see if all the reviewers are prepared. This is done by a brief examination of the effort and defect data in the selfreview logs to confirm that sufficient time and attention has gone into the preparation. When preparation is not adequate, the group review is deferred until all participants are fully prepared. If everything is ready, the group review meeting is held. The moderator is incharge of the meeting and has to make sure that the meeting stays focused on its basic purpose of defect identification and does not degenerate into a general brainstorming session or personal attacks on the author. The meeting is conducted as follows.

A team member (called the reader) goes over the work product line by line (or any other convenient small unit), and paraphrases each line to the team. Sometimes no paraphrasing is done and the team just goes over the work product line by line. At any line, if any reviewer has any issue from before, or finds any new issue in the meeting while listening to others, the reviewer raises the issue. There could be a discussion on the issue raised. The author accepts the issue as a defect or clarifies why it is not a defect. After discussion an agreement is reached and one member of the review team (called the scribe) records the identified defects in the defect log. At the end of the meeting, the scribe reads out the defects recorded in the defect log for a final review by the team members. Note that during the entire

process of review, defects are only identified. It is not the purpose of the group to identify solutions—

Project	Xxxxxxxx
Work Product Type	Project Plan, V 1.0
Size of Product	14 pages
Review Team	P1, P2, P3, P4
Effort (Person Hours)	
Preparation	Total 10 person-hrs.
Group Review Meeting	10 person-hrs.
Total Effort	**20 person-hrs.**
Defects	
Number of Critical Defects	0
Number of Major Defects	3
Number of Minor Defects	16
Total Number of defects	**19**
Review Status	Accepted
Recommendations for next phase	
Comments	The plan has been well documented and presented

Figure 2.14: Summary report of an inspection.

that is done later by the author.

The final defect log is the official record of the defects identified in the inspection and may also be used to track the defects to closure. For analyzing the effectiveness of a review, however, only summary level information is needed, for which a summary report is prepared. The summary report describes the work product, the total effort spent and its breakup in the different review process activities, total number of defects found for each category, and size. If types of defects were also recorded, then the number of defects in each category can also be recorded in the summary.

The summary report is self-explanatory. Total number of minor defects found was 19, and the total number of major defects found was 3. That is, the defect density found is 16/14 = 1.2 minor defects per page, and 3/14 — 0.2 major defects per page. From experience, both of these rates are within the range seen in the past; hence it can be assumed that the review was conducted properly. The review team had 4 members, and each had spent 2.5 hours in individual review and the review meeting lasted 2.5 hours. This means that the coverage rate during preparation and review was 14/2.5 = 5.6 pages per hour, which, from past experience, also seems acceptable. If the modifications required for fixing the defects and addressing the issues are few, then the group review status is "accepted." If the modifications required are many, a follow up meeting by the moderator or a re-review might be necessary to verify whether the changes have been incorporated correctly. The moderator recommends what is to be done. In addition, recommendations regarding reviews in the next stages may also be made (e.g., in a detail design review it may be recommended code of which modules should undergo inspections.)

Rework and Follow- Up

In this phase the author corrects all the defects raised during the inspection. The author may redo the work product, if that is what the moderator recommended. The author reviews the corrections with the moderator or in a re-review, depending on the decision of the group review meeting. The scribe ensures that the group review report and minutes of the meetings are communicated to the group review team.

Roles and Responsibilities

The inspection process is a structured process with different people having different responsibihties. The key roles in a group review are those of moderator, reader, scribe, author, and reviewer. These are logical roles and a person can be assigned multiple roles, with the restrictions that the author cannot be the moderator or the reader, and the moderator cannot be the reader. This implies that the minimum size of the group review team is three—the author, the moderator, and the reader. These three people are also reviewers and can assign the role of scribe to someone. The responsibilities of these roles should be clear from the inspection process. Here we briefly summarize the main activities of the

moderator and the reviewers. The moderator perhaps has the most important role during a group review. He has the overall responsibility of ensuring that the review goes well. The moderator should undergo formal training on how to conduct reviews, or should have experience of participating in a few reviews. The responsibilities of the moderator include:

• Schedule the group review meeting

• At the opening of group review meeting ensure that all participants are prepared and have submitted self-preparation log, or reschedule the group review

• Conduct the group review in an orderly and efficient manner

• Ensure that the meeting stays focused on the main task of defect identification

• Track each problem to resolution or ensure that it is tracked by someone else

• Ensure that group review reports are completed

During the meeting, the moderator has to make sure that all the participants contribute effectively, everyone is heard, there is an agreement on the findings of the review, and that the interest level does not drop. A key responsibility is to ensure that during the meeting the focus remains on problem identification and does not drift into problem resolution and that all reviewers remain focused on finding defects in the work product and do not get into finding faults with the author. Overall, orderly and amicable conduct of the meeting is largely the responsibility of the moderator. After the meeting, the moderator has to make sure that all participants are satisfied, the review reports have been filled and follow-up actions taken. A reviewer is primarily responsible for finding defects. Generally, all members of the group review team are reviewers. The defects are found either through individual review or through the group review meeting. The main issues for a reviewer are:

• Be prepared for group review

• Be objective; focus on issues and not on people

• Concentrate on problems (offer solutions only after the group review)

• If something is not clear do not hesitate to stop progress until it is understood

• When proved wrong, move on

Guidelines for Work Products

All the work products in a project may not undergo group review as that may be prohibitively expensive and may not give commensurate returns. For each project it has to be decided which work products should be inspected, and the size of the inspection team. As the work products of the early part of the life cycle are very critical and defects in them have a multiplier effect in the later stages, it is recommended that early work products like the requirements document, architecture document, and project management plan, be inspected. Regarding team size, though a team size of three to five is often recommended, sometimes where the cost is not justified, an inspection team of just the author and another reviewer may be suitable. This is also sometimes called one-person review. Though the inspection process is same for any work product, the focus of the inspection is often different for different products. The constitution of the review team and the checklists used in review also depend on the nature of the work product. Some of the guidelines regarding the focus of the review and the composition of the inspection team are given in Table.

Work product	Focus of Inspection	Participants
Requirement Specification	Requirements meet customer needs Requirements are implementable Omissions, inconsistencies and ambiguities in the requirements	Customer Designers Tester Developer
High Level Design	High-level design implements the requirements The design is implementable Omissions, and other defects in the design	Requirements author Detailed designer Developer
Code	Code implements the design Code is complete and correct Defects in code	Designer Tester Developer
System Test Cases	The set of test cases checks all conditions in the requirements Test cases are executable	Requirements author Tester Project leader
Project Management Plan	Plan is complete Project management plans is implementable Omissions and ambiguities	Project leader SEPG member Another project leader

It is often hard to believe that a human-intensive process like the inspections can improve quality and productivity. Due to this and other reasons, inspections are often resisted.

Reference:

1. Pressman: Software Engineering, Tata McGraw Hill.

2. Sommerville, I., 1986: Software Engineering, Narosa Publ. House.

3. Mall,Rajib, 2009: Fundamentals of Software Engineering.

4. Jalote, Pankaj, 1995: An Integrated Approach to Software Engineering, Narosa Publ